

BackVigenere CipherGraded Quiz + 25 min

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Instructor Introduction and Module Objectives

Alphabet

Substitution Cipher

Modulo Operation

Monoalphabetic Cipher

Natural Language Redundancy

Polyalphabetic Cipher

Video: Polyalphabetic Cipher and Vigenere Cipher8 min

Reading: Lecture Slides for Classical Cipher - Substitution15 min

Quiz: Vigenere CipherSubmitted

Pseudo-Code: Vigenere Cipher

● Congratulations! You passed!

Grade received100%

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Vigenere Cipher

Review Learning Objectives

1. Let's use Vigenere Cipher to encrypt English text, ignoring capitalization and other symbols beyond English alphabets. The key is of length 1 and of value "Y" or 24, i.e., the key is "Y". Which of the following statements is **false**?

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Try again

The cipher can be expressed as $c = (p+24) \bmod 26$, where c is each ciphertext letter and p is each plaintext letter.

The cipher is equivalent to the Vigenere cipher using a key of length 3 and of value "YYY".

The cipher can be expressed as $c = (p-2) \bmod 26$, where c is each ciphertext letter and p is each plaintext letter.

To Pass70% or higher

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☐

The cipher is equivalent to a Caesar Cipher.

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Correct

See Polyalphabetic Cipher and Vigenere Cipher lesson.

2. Vigenere cipher is used with a key "KQ". What is the ciphertext corresponding to the plaintext, DRAGON?

4 / 4 points

AFKJLLB

☒

Correct

See Polyalphabetic Cipher and Vigenere Cipher lesson.

3. Vigenere cipher is used with a key "KQ" to encrypt a long English book, comprised of English alphabet letters and ignoring capitalization and other symbols. The key length is fixed, i.e., the key is 2 alphabets long. What is the number of possible distinct keys?

4 / 4 points

676

☒

Correct

See Polyalphabetic Cipher and Vigenere Cipher lesson.

4. Vigenere cipher is used with a key "KQ" to encrypt a long English book. You order the letters appearing in the encrypted book in the decreasing order of frequency (i.e., from the most frequently appearing to the least frequently appearing), and the result is: RGZNFIBJWPESCNITWUJLQDKQ. Which ciphertext letters correspond to the plaintext letter E (which is the most frequent letter in the plaintext)? Apply all those that apply.

6 / 6 points

☐

Z

☐

T

☐

E

☐

A

☒

R

☒

Correct

The most frequent ciphertext letter likely corresponds to the most frequent plaintext letter.

☐

Y

☒

G

☒

Correct

The key is of length 2, so one plaintext letter can get mapped to two distinct ciphertext letters.

https://www.coursera.org/learn/symmetric-crypto/exam/CPmJL/Vigenere-cipher/attempt/redToCover>true

1/1